

# Breeti Bandyopadhyay

88904 72117 | [bandybreeti@gmail.com](mailto:bandybreeti@gmail.com) | [linkedin.com/in/breeti](https://www.linkedin.com/in/breeti) | [silvermysti.github.io](https://silvermysti.github.io)

## EDUCATION

### Shiv Nadar University

*B.Tech Computer Science and Engineering*

- CGPA: 9.25 · Pursuing a minor in Mathematics.
- Dean's List (2024/25): ranked in the top 10% of the Computer Science cohort.

### Birla Balika Vidyapeeth

*Secondary and Senior Secondary (CBSE)*

- Class X: 97.6% | Class XII: 97%

Greater Noida, UP

2024 – 2028

Pilani, Rajasthan

2021 – 2024

## EXPERIENCE

### Full-Stack Dev Intern

May 2026 – Jul 2026

*Tech Mahindra*

*NSEZ Noida*

- Engineered a multi-agent LLM system for automotive warranty claims: structured-output (Pydantic function-calling) decisioning across intake, coverage, fraud, and recommendation agents on a LangGraph pipeline with human-in-the-loop approval.
- Built a bounded conversational intake agent with conditional vision-evidence requests; tuned prompts and structured decision logic across fast/standard model tiers (Groq / DeepSeek).

### Research Intern

Jun 2025 – Jul 2025

*5G Use Case Lab, BITS Pilani*

*Pilani, Rajasthan*

- Replicated a peer-reviewed IEEE paper end-to-end: VGG-style 1D CNN for automatic modulation classification on 80,000 examples from a 19.5 GB dataset (DeepSig RadioML 2018.01A), trained on CPU.
- Engineered a confidence-based abstention gate lifting effective accuracy to 90% on answered signals; validated using blind SNR estimation ( $r = -0.996$ ) with no ground-truth labels.

## RESEARCH

### Research Collaborator

Dec 2024 – Jan 2025

*Physics Department, BITS Pilani*

*Pilani, Rajasthan*

- Co-authored a DFT study on reversible hydrogen storage in 2D holey graphyne, published in *Int. J. Hydrogen Energy* (Impact Factor: 8.3); DOI: [10.1016/j.ijhydene.2025.150044](https://doi.org/10.1016/j.ijhydene.2025.150044).
- Conducted VASP simulations for structure optimisation and electronic analysis (DOS, band structure); evaluated adsorption metrics meeting the DOE target.

## PROJECTS

### Hephaestus | Iterative Code Performance Optimiser

- Profiles binaries with Linux `perf`, feeds hardware counters (IPC, cache misses, branch mispredictions) to an LLM, and applies one optimisation at a time in a loop until code reaches within 5% of theoretical CPU peak.
- Weighted scoring (70% runtime / 20% IPC / 10% cache miss rate) auto-rolls back ineffective changes; supports x86\_64, arm64, wasm32, riscv64 via Docker with a Go HTTP server + Python perf-agent backend and SvelteKit + Supabase frontend.

### Warranty Claims Decision Engine | Multi-agent LLM Pipeline

- Automates warranty-claim decisions with a pipeline of agents that reads the complaint, checks coverage, estimates repair cost, and screens for fraud, escalating risky or high-value claims to a human manager.
- Payouts are kept auditable by computing coverage and cost in plain code against a parts-and-policy database; the model only classifies and explains, so routine claims clear automatically.
- Stack:** Python, FastAPI, LangGraph, SQLAlchemy, Next.js / TypeScript, Groq / DeepSeek

## COURSES & CERTIFICATIONS

**LinkedIn CoachIn Program (2026):** Selected among the top 100 women engineering students nationwide for intensive mentorship in DSA, problem-solving, and tech career preparation.

**Machine Learning Specialization** (DeepLearning.AI & Stanford University): supervised learning, advanced learning algorithms, unsupervised learning, recommenders, and reinforcement learning.

**AWS Academy Graduate:** Cloud Architecting · Cloud Foundations. **IBM:** Introduction to Software Engineering.

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, Go, TypeScript / JavaScript, SQL, R

**AI & Backend:** LangGraph, LLM agent workflows, FastAPI, Flask, SQLAlchemy, Pydantic, REST APIs

**ML / CV:** TensorFlow, Keras, Scikit-Learn, XGBoost, NumPy, pandas, OpenCV, MediaPipe

**Web & Data:** React, Next.js, Svelte / SvelteKit, Tailwind CSS · MySQL, PostgreSQL, SQLite, Supabase

**Tools:** Git, Docker, Linux, Linux `perf`, Jupyter, VS Code, VASP, VESTA

## HONOURS & ACHIEVEMENTS

**State Topper, Rajasthan:** KAMP NASTA 2023 — 1st in state out of 500,000+ participants.

**HackData SNU:** 3rd place, Shiv Nadar University's flagship national-level hackathon.

**School Rank 2:** Class XII (CBSE), Birla Balika Vidyapeeth.